

1)

$$1. D_{KL}(P_{data}(x|y) || P_{\theta}(x|y)) = \mathbb{E}_{x \sim P_{data}} \left[\log \left(\frac{P_{data}(x|y)}{P_{\theta}(x|y)} \right) \right]$$

$$= \mathbb{E}_{x, y \sim P_{data}} [\log P_{data}(x|y)] - \mathbb{E}_{x, y \sim P_{data}} [P_{\theta}(x|y)]$$

$$2. P_{\theta}(x|y) = \int P_{\theta}(x|y, z) P_{\theta}(z|y) = \mathbb{E}_{z \sim P_{\theta}(z|y)} [P_{\theta}(x|y, z)]$$

$$3. P_{\theta}(x|y) = \int \frac{P_{\theta}(z|y)}{q(z|x, y)} P_{\theta}(x|y, z) q(z|x, y) = \mathbb{E}_{z \sim q(z|x, y)} \left[\underbrace{\frac{P_{\theta}(z|y)}{q(z|x, y)} P_{\theta}(x|y, z)}_A \right]$$

$$4. \log P_{\theta}(x|y) = \log \mathbb{E}_{z \sim q(z|x, y)} \left[\frac{P_{\theta}(z|y)}{q(z|x, y)} P_{\theta}(x|y, z) \right]$$

$$\geq \mathbb{E}_{z \sim q(z|x, y)} [\log P_{\theta}(z|y) + \log P_{\theta}(x|y, z) - \log(q(z|x, y))]$$

$$= \mathbb{E}_{z \sim q(z|x, y)} [\underbrace{\log P_{\theta}(z|y) + \log P_{\theta}(x|y, z)}_{\log P_{\theta}(x, z|y)}] + H(q(z|x, y)) = ELBO(x, y; \theta, q)$$

5.

$$D_{KL}(q(z|x, y) || P_{\theta}(z|y)) = \mathbb{E}_{z \sim q(z|x, y)} [\log q(z|x, y) - \log P_{\theta}(z|y)]$$

$$= -H(q(z|x, y)) - \mathbb{E}_{z \sim q(z|x, y)} [\log P_{\theta}(x, z|y) - \log P_{\theta}(x|y)]$$

$$= \log P_{\theta}(x|y) - ELBO(x, y; \theta, q)$$

$$\Rightarrow D_{KL}(q(z|x, y) || P_{\theta}(z|x, y)) + ELBO(x, y; \theta, q) = \log P_{\theta}(x|y)$$

$$6. q^*_{z|x, y} = \arg \min_q D_{KL}(q || P_{\theta}(z|x, y))$$

$$P_{\theta}(z|x, y) = \frac{P_{\theta}(z, x|y)}{P_{\theta}(x|y)}$$

we don't have this!

→ minimizing D_{KL} is intractable

because we don't have $P_{\theta}(z|x, y)$

Sol. $\log p_{\theta}(x|y) = \text{ELBO}(x, y; \theta, q) + D_{KL}(q \| p_{\theta}(z|y))$

$$\Rightarrow q^* = \underset{q}{\operatorname{argmin}} D_{KL}(q \| p_{\theta}(z|y)) = \underset{q}{\operatorname{argmax}} \text{ELBO}(x, y; \theta, q)$$

$\Rightarrow$ we optimize ELBO w.r.t. q

$$q \sim \mathcal{N}(\mu, \sigma^2 I) \Rightarrow z = \mu + \sigma \epsilon \sim q \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\begin{aligned} \nabla_{\lambda} \text{ELBO} &= \nabla_{\lambda} \mathbb{E}_{z \sim q(z|x, y)} [\log p_{\theta}(z|y) + \log p_{\theta}(x|y, z) - \log(q(z|x, y))] \\ &= \nabla_{\lambda} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [\log p_{\theta}(\underbrace{\mu + \sigma \epsilon}_{g_{\lambda}(\epsilon)}) + \log p_{\theta}(x|y, \mu + \sigma \epsilon) - \log(q(\mu + \sigma \epsilon|x, y))] \\ &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} [\nabla_{\lambda} \log p_{\theta}(z, g_{\lambda}(\epsilon)|y) - \nabla_{\lambda} \log(q(g_{\lambda}(\epsilon)|x, y))] \end{aligned}$$

we optimize this using MC and gradient descent.

7. calculating expectations using Monte-Carlo estimation.

$$\begin{aligned} \text{ELBO}(x_{\text{new}}, y_{\text{new}}; \theta, q) &\approx \frac{1}{N} \sum_{i=1}^N \log p_{\theta}(x_{\text{new}}|z_i, y_{\text{new}}) - \log p_{\theta}(z_i|y_{\text{new}}) \\ &\quad - \log q(z_i|x_{\text{new}}, y_{\text{new}}) \quad z_i \sim q(z_i|x_{\text{new}}, y_{\text{new}}) \end{aligned}$$

$$p_{\theta}(x_{\text{new}}|y_{\text{new}}) = \mathbb{E}_{p_{\theta}(z|y)} [p_{\theta}(x_{\text{new}}|y_{\text{new}}, z)] = \mathbb{E}_{q(z|x, y)} \left[\frac{p_{\theta}(z|y_{\text{new}})}{q(z|x_{\text{new}}, y_{\text{new}})} p_{\theta}(x_{\text{new}}|y_{\text{new}}, z) \right]$$

$$\Rightarrow p_{\theta}(x_{\text{new}}|y_{\text{new}}) \approx \frac{1}{N} \sum_{i=1}^N \frac{p_{\theta}(x_{\text{new}}|y_{\text{new}}, z_i) p_{\theta}(z_i|y_{\text{new}})}{q(z_i|x_{\text{new}}, y_{\text{new}})} \quad z_i \sim q(z_i|x_{\text{new}}, y_{\text{new}})$$

we calculate $p_{\theta}(x|y)$ using importance sampling to reduce variance.

we then use Monte-Carlo estimation.

8. y_{new}

$$\tilde{x} = \underset{x}{\operatorname{argmax}} p_{\theta}(x|y_{\text{new}}) \approx \underset{x, z}{\operatorname{argmax}} p_{\theta}(x|y_{\text{new}}, z) q(z|x, y_{\text{new}})$$

to generate:

$$\tilde{z} \sim q(z|\text{None}, y_{\text{new}}) \rightarrow \tilde{x} \sim p_{\theta}(x|y_{\text{new}}, \tilde{z})$$

2)

$$1. \quad p \sim \mathcal{N}(\mu_1, \sigma_1^2)$$

$$q \sim \mathcal{N}(\mu_2, \sigma_2^2)$$

$$D_{\text{cs}}(P \parallel Q) = -\log \frac{\int p(x) q(x) dx}{\sqrt{\int p(x)^2 dx \int q(x)^2 dx}} = \frac{1}{2} \log \int p(x)^2 dx + \frac{1}{2} \log \int q(x)^2 dx - \log \int p(x) q(x) dx$$

$$\int p(x) q(x) dx = \int_{\mathbb{R}} \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(x-\mu_2)^2}{2\sigma_2^2}\right) dx$$

$$= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(x^2 + \mu_1^2 - 2x\mu_1)\sigma_2^2 + (x^2 + \mu_2^2 - 2x\mu_2)\sigma_1^2}{2\sigma_1^2\sigma_2^2}\right) dx$$

$$\sigma^2 = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

$$\begin{aligned} \mu &= \frac{\mu_1\sigma_2^2 + \mu_2\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \Rightarrow \frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(x-\mu_2)^2}{\sigma_2^2} = x^2\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right) + \frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} \\ &\quad - 2x\left(\frac{\mu_1}{\sigma_1^2} + \frac{\mu_2}{\sigma_2^2}\right) \\ &= \frac{x^2}{\sigma^2} + \frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - 2x \frac{\mu(\sigma_1^2 + \sigma_2^2)}{\sigma_1^2\sigma_2^2} \\ &= \frac{x^2}{\sigma^2} - \frac{2x\mu}{\sigma^2} + \frac{\mu^2}{\sigma^2} + \left(\frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - \frac{\mu^2}{\sigma^2}\right) \end{aligned}$$

$$\Rightarrow \int p(x) q(x) dx = \int_{\mathbb{R}} \frac{1}{2\pi\sigma_1\sigma_2} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2} - \frac{1}{2}\left(\frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - \frac{\mu^2}{\sigma^2}\right)\right] dx$$

$$= \frac{\sigma}{\sqrt{2\pi}\sigma_1\sigma_2} \exp\left(-\frac{1}{2}\left(\frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - \frac{\mu^2}{\sigma^2}\right)\right)$$

$$\Rightarrow \int_{\mathbb{R}} p(x)^2 dx = \frac{1}{\sqrt{2\pi}\sqrt{2}\sigma_1} \quad \int_{\mathbb{R}} q(x)^2 dx = \frac{1}{\sqrt{2\pi}\sqrt{2}\sigma_2}$$

$$\Rightarrow D_{\text{cs}}(P \parallel Q) = -\log\left[\frac{\sqrt{2}\sigma}{\sqrt{\sigma_1\sigma_2}} \exp\left(-\frac{1}{2}\left(\frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - \frac{\mu^2}{\sigma^2}\right)\right)\right]$$

$$\Rightarrow D_{cs}(P||Q) = -\log \frac{\sqrt{2}\sigma}{\sqrt{\sigma_1\sigma_2}} + \frac{1}{2} \left(\frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - \frac{\mu^2}{\sigma^2} \right)$$

$$D_{KL}(P||Q) = \int_{\mathbb{R}} \log \frac{P(x)}{Q(x)} P(x) dx$$

$$= \int_{\mathbb{R}} \left[\log \left(\frac{\sigma_2}{\sigma_1} \right) + \left(\frac{(x-\mu_2)^2}{2\sigma_2^2} - \frac{(x-\mu_1)^2}{2\sigma_1^2} \right) \right] P(x) dx$$

$$= \log \left(\frac{\sigma_2}{\sigma_1} \right) - \frac{1}{2} + \frac{1}{2} \int_{\mathbb{R}} \frac{x^2 + \mu_2^2 - 2x\mu_2}{\sigma_2^2} P(x) dx$$

$$= \log \left(\frac{\sigma_2}{\sigma_1} \right) - \frac{1}{2} + \frac{1}{2} \left(\frac{\sigma_1^2 - \mu_1^2 + \mu_2^2 - 2\mu_1\mu_2}{\sigma_2^2} \right)$$

$$= -\frac{1}{2} + \log \left(\frac{\sigma_2}{\sigma_1} \right) + \frac{1}{2} \frac{\sigma_1^2}{\sigma_2^2} + \frac{(\mu_2 - \mu_1)^2}{2\sigma_2^2} = \frac{1}{2} \left(\log \left(\frac{\sigma_1^2}{\sigma_2^2} \right) + \frac{\sigma_1^2}{\sigma_2^2} + \left(\frac{\mu_2 - \mu_1}{\sigma_2} \right)^2 - 1 \right)$$

$$D_{KL}(Q||P) = \frac{1}{2} \left(\log \left(\frac{\sigma_1^2}{\sigma_2^2} \right) + \frac{\sigma_2^2}{\sigma_1^2} + \left(\frac{\mu_1 - \mu_2}{\sigma_1} \right)^2 - 1 \right)$$

$$D_{cs}(P||Q) = -\frac{1}{2} \log \frac{2\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2} + \frac{1}{2} \left(\frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} - \frac{\mu^2}{\sigma^2} \right)$$

$$\mu^2 = \frac{1}{(\sigma_1^2 + \sigma_2^2)^2} (\mu_1^2 \sigma_2^4 + \mu_2^2 \sigma_1^4 + 2\mu_1\mu_2 \sigma_1^2 \sigma_2^2) \Rightarrow \frac{\mu^2}{\sigma^2} = \frac{1}{\sigma_1^2 \sigma_2^2 (\sigma_1^2 + \sigma_2^2)}$$

$$(\mu_1^2 \sigma_2^4 + \mu_2^2 \sigma_1^4 + 2\mu_1\mu_2 \sigma_1^2 \sigma_2^2) =$$

$$R = \frac{1}{\sigma_1^2 \sigma_2^2 (\sigma_1^2 + \sigma_2^2)} (\mu_1^2 \sigma_2^2 (\sigma_1^2 + \sigma_2^2) + \mu_2^2 \sigma_1^2 (\sigma_1^2 + \sigma_2^2) - \mu_1^2 \sigma_2^4 - \mu_2^2 \sigma_1^4 - 2\mu_1\mu_2 \sigma_1^2 \sigma_2^2)$$

$$= \frac{1}{\sigma_1^2 + \sigma_2^2} (\mu_1^2 + \mu_2^2 - 2\mu_1\mu_2) = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$$

$$D_{cs}(P||Q) = -\frac{1}{2} \log \frac{2\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2} + \frac{1}{2} \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$$

$$D_{KL}(P||Q) - D_{cs}(P||Q) = \frac{1}{2} \left(\log \left(\frac{\sigma_1^2}{\sigma_2^2} \right) + \frac{\sigma_1^2}{\sigma_2^2} + \left(\frac{\mu_2 - \mu_1}{\sigma_2} \right)^2 - 1 \right) - \frac{1}{2} \log \left(\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2} \right)$$

$$(\mu_1 - \mu_2)^2 \left(\frac{1}{\sigma_2^2} - \frac{1}{\sigma_1^2 + \sigma_2^2} \right) \geq 0$$

$$\Delta(u) = \log \left(\frac{\sigma_1^2}{\sigma_2^2} \right) + \frac{\sigma_1^2}{\sigma_2^2} - 1 - \log \left(\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2} \right) \Rightarrow \frac{\partial}{\partial u} \Delta(u) = \frac{-2}{u} + 2u - \frac{2u}{u^2 + 1} + \frac{2}{2u}$$

$$= -\frac{2}{u} + 2u - \frac{2u}{u^2+1} + \frac{1}{u} = \frac{2u^3}{u^2+1} - \frac{1}{u} = 0 \Rightarrow 2u^4 - u^2 - 1 = 0 \Rightarrow u^2 = \frac{1}{4} \pm \frac{3}{4}$$

$$\Rightarrow u^2 = 1 \Rightarrow u^* = \pm 1$$

$$\Rightarrow \Delta(u^*) = \log(1) + 1 - 1 - \log\left(\frac{2}{2}\right) = 0$$

$$\Rightarrow \Delta(u) \geq 0 \Rightarrow D_{KL}(P||Q) \geq D_{CS}(P||Q)$$

$$\text{likewise: } D_{KL}(Q||P) \geq D_{CS}(P||Q)$$

$$\Rightarrow D_{CS}(P||Q) \leq \min(D_{KL}(P||Q), D_{KL}(Q||P))$$

□

2.

$$L_{CS}(x; \theta, \phi) = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \lambda D_{CS}(q_{\phi}(z|x) || p(z))$$

$$\log p_{\theta}(x) = D_{KL}(q_{\phi}(z|x) || p_{\theta}(z|x)) + ELBO(x; \theta, \phi)$$

$$ELBO(x; \theta, \phi) = \mathbb{E}_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(x|z)] + \mathbb{E}_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(z)] - \mathbb{E}_{z \sim q_{\phi}(z|x)} [\log q_{\phi}(z|x)]$$

$$= \mathbb{E}_{z \sim q_{\phi}} [\log p_{\theta}(x|z)] - D_{KL}(q_{\phi}(z|x) || p_{\theta}(z))$$

$$\Rightarrow L_{CS}(x; \theta, \phi) = ELBO(x; \theta, \phi) + D_{KL}(q_{\phi}(z|x) || p_{\theta}(z)) - \lambda D_{CS}(q_{\phi}(z|x) || p_{\theta}(z))$$

$$= \log p_{\theta}(x) - D_{KL}(q_{\phi}(z|x) || p_{\theta}(z|x)) + D_{KL}(q_{\phi}(z|x) || p_{\theta}(z)) - \lambda D_{CS}(q_{\phi}(z|x) || p_{\theta}(z))$$

This loss maximizes ELBO like VAE and also holding a more complex regularization term.

- if λ is too high the encoder learns nothing but the prior and the quality decreases.

- if λ is too small may lead to mode collapse as in traditional VAE.

3)

1. posterior collapse is a phenomenon where the posterior of latents is equal to its prior. The encoder distribution ($q_{\phi}(z|x)$) is indistinguishable from the prior $p(z)$.

encoder produces same generic summary of any given inputs and its encoded latent contains no useful information.

VAE:

$$\max_{\theta, \phi} \mathbb{E}_{z \sim q_{\phi}} (\log p_{\theta}(x|z)) - D_{KL}(q_{\phi}(z|x) || p(z))$$

the optimization process finds a trivial solution where:

$$q_{\phi}(z|x) \approx p(z), \quad D_{KL}(q_{\phi}(z|x) || p(z)) \approx 0$$

2.

$$\max_{\theta, \phi} ELBO(x; \theta, \phi) = \mathbb{E}_{z \sim q_{\phi}} (\log p_{\theta}(x|z)) - D_{KL}(q_{\phi}(z|x) || p(z))$$

$$< \log p_{\theta}(x)$$

$$p_{\theta}^*(x|z) \approx p_{data}(x) \Rightarrow \max_{\theta, \phi} ELBO(x; \theta, \phi) =$$

$$\max_{\theta, \phi} \underbrace{\mathbb{E}_{z \sim q_{\phi}} (\log p_{data}(x))}_{\log p_{data}(x)} - D_{KL}(q_{\phi} || p)$$

$$= \log p_{data}(x) - \min_{\phi} D_{KL}(q_{\phi}(z|x) || p(z))$$

$$\Rightarrow q_{\phi}^*(z|x) = p(z) \quad \square$$

3.

$$\begin{aligned} \text{ELBO}(\theta, \phi) &= \mathbb{E}_{x \sim p_{\text{data}}} [\text{ELBO}(x; \theta, \phi)] \\ &= \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{z \sim q_{\phi}} [\log p_{\theta}(x|z)] - \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{z \sim q_{\phi}} \left[\log \frac{q_{\phi}(z|x)}{p(z)} \right] \end{aligned}$$

$$\mathbb{E}_{x \sim p_{\text{data}}} \left[\int \log \left(\frac{q_{\phi}(z|x)}{p(z)} \right) q_{\phi}(z|x) dz \right] =$$

$$\iint p_{\text{data}}(x) \cdot \log \left(\frac{q_{\phi}(z|x)}{p(z)} \right) q_{\phi}(z|x) dz dx =$$

$$\frac{q_{\phi}(z|x)}{q_{\phi}(z)} \cdot \frac{q_{\phi}(z)}{p(z)}$$

$$\int \log \left(\frac{q_{\phi}(z)}{p(z)} \right) \left(\int p_{\text{data}}(x) q_{\phi}(z|x) dx \right) dz$$

$$\iint p_{\text{data}}(x) q_{\phi}(z|x) \log \left(\frac{q_{\phi}(z|x)}{q_{\phi}(z)} \right) dz dx + \iint p_{\text{data}}(x) q_{\phi}(z|x) \log \left(\frac{q_{\phi}(z)}{p(z)} \right) dz dx$$

$$= \underbrace{\mathbb{E}_{x \sim p_{\text{data}}} [D_{\text{KL}}(q_{\phi}(z|x) || q_{\phi}(z))]}_{I_f(x; z)} + D_{\text{KL}}(q_{\phi}(z) || p(z))$$

$I_f(x; z)$

$$\Rightarrow \text{ELBO}(\theta, \phi) = \mathbb{E}_{x \sim p_{\text{data}}} \mathbb{E}_{z \sim q_{\phi}} [\log p_{\theta}(x|z)] - D_{\text{KL}}(q_{\phi}(z) || p(z)) - I_f(x; z)$$

4.

1) Decoder is too powerful

optimization relies on decoder and encoder receives no grad signal to encode meaningful information.

2) The prior biases us.

$I_f(x; z)$ penalizes the storage of information in encoder.

In the early stages the prior forces the model to converge to a local minimum with no information in z .

3) order of param optimization.

encoder needs decoder to use z and produce feedback for it.

at early stages decoder doesn't know how to use z and with meaningless z , it stays that way. $\rightarrow$ no signal for encoder

5.

1) Beta VAE. $\mathcal{L} = \mathbb{E}_{\mathbf{z}} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] - \beta D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) || p_{\theta}(\mathbf{z}))$

By scheduling β (starting at 0) we can induce encoder to learn info and then force it to hold near prior.

2) Updating encoder more aggressively.

train encoder for k steps on decoder for 1 step each time.

encoder learns to encode info, and then the decoder receives useful info it can't ignore.

3) Threshold KL to prior. $\mathcal{L} = \mathbb{E}_{\mathbf{z}} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] - \max(1, D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) || p_{\theta}(\mathbf{z})))$

if $D_{KL} \leq 1$, the gradient of KL w.r.t. encoder becomes zero.

this removes the pressure on posterior to collapse into prior.

4)

1.

$$\begin{aligned}
p_{\theta}(X_{1:T}, A_{1:T}) &= \mathbb{E}_{p_{\theta}(z_{1:T})} [p_{\theta}(X_{1:T}, A_{1:T} | z_{1:T})] \\
&= \mathbb{E}_{q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T})} \left[\frac{p_{\theta}(z_{1:T})}{q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T})} p_{\theta}(X_{1:T}, A_{1:T} | z_{1:T}) \right] \\
\Rightarrow \log p_{\theta}(X_{1:T}, A_{1:T}) &= \log \mathbb{E}_{q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T})} \left[\frac{p_{\theta}(z_{1:T})}{q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T})} p_{\theta}(X_{1:T}, A_{1:T} | z_{1:T}) \right] \\
&\geq \mathbb{E}_{q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T})} [\log(p_{\theta}(X_{1:T}, A_{1:T}, z_{1:T}) - \log q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T}))] \\
&= \mathbb{E}_{q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T})} [\log(p_{\theta}(X_{1:T}, A_{1:T} | z_{1:T}))] \left. \begin{array}{l} \text{ELBO}(X_{1:T}, A_{1:T}; \theta, \phi) \\ - D_{KL}(q_{\phi}(z_{1:T} | X_{1:T}, A_{1:T}) || p_{\theta}(z_{1:T})) \end{array} \right\}
\end{aligned}$$

$$\begin{aligned}
\text{ELBO}(X_{1:T}, A_{1:T}; \theta, \phi) &= \mathbb{E}_{q_{\phi}(z_t | X_{1:t}, A_{1:t})} \left[\sum_{t=1}^T \log p_{\theta}(X_t, A_t | z_{t-1}) \right] \\
&\quad - \mathbb{E}_{q_{\phi}(z_t | X_{1:t}, A_{1:t})} \left[\sum_{t=1}^T \log(q_{\phi}(z_t | X_{1:t}, A_{1:t})) \right] \\
&\quad + \mathbb{E}_{z_t \sim q_{\phi}} [\log(p(z_0) p_{\theta}(z_1 | z_0) \dots p_{\theta}(z_T | z_{T-1}))] \\
&\quad \quad \quad \downarrow z_0 \text{ is assumed fixed.} \\
&= \underbrace{\mathbb{E}_{z_t \sim q_{\phi}(z_t | X_{1:t}, A_{1:t})} \left[\sum_{t=1}^T \log p_{\theta}(X_t, A_t | z_{t-1}) \right]}_{\text{reconstruction}} - \underbrace{\mathbb{E}_{z_t \sim q_{\phi}(z_t | X_{1:t}, A_{1:t})} \left[\sum_{t=1}^T D_{KL}(q_{\phi}(z_t | X_{1:t}, A_{1:t}) || p_{\theta}(z_t | z_{t-1})) \right]}_{\text{KL}}
\end{aligned}$$

2.

$$\log p_{\theta}(X_t, A_t | z_{t-1}) = \log p_{\theta}(X_t | z_{t-1}) + \log p_{\theta}(A_t | z_{t-1})$$

$$p_{\theta}(X_t | z_{t-1}) = \frac{1}{\prod_{v=1}^n \sqrt{(2\pi)^d \det(\Sigma_{\theta}^v(z_{t-1}))}} \exp\left(-\frac{1}{2} (x_t^v - \mu_{\theta}^v(z_{t-1}))^T \Sigma_{\theta}^{-1}(z_{t-1}) (x_t^v - \mu_{\theta}^v(z_{t-1}))\right)$$

$$\Rightarrow \log p_{\theta}(X_t | z_{t-1}) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \sum_{v=1}^n \log(\det \Sigma_{\theta}^v(z_{t-1}))$$

$$- \frac{1}{2} (X_t - \mu_{\theta}^v(z_{t-1}))^T \Sigma_{\theta}^{v^{-1}}(z_{t-1}) (X_t - \mu_{\theta}^v(z_{t-1}))$$

$$p_{\theta}(A_t | z_{t-1}) = \prod_{u < v} [\pi_{\theta}^{uv}(z_{t-1})]^{A_t^{uv}} [1 - \pi_{\theta}^{uv}(z_{t-1})]^{1-A_t^{uv}}$$

$$\Rightarrow \log p_{\theta}(A_t | z_{t-1}) = \sum_{u < v} A_t^{uv} \log \pi_{\theta}^{uv}(z_{t-1}) + (1-A_t^{uv}) \log(1 - \pi_{\theta}^{uv}(z_{t-1}))$$

$$\Rightarrow \log p_{\theta}(X_t, A_t | z_{t-1}) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \sum_{v=1}^n \log(\det \Sigma_{\theta}^v(z_{t-1}))$$

$$- \frac{1}{2} (X_t - \mu_{\theta}^v(z_{t-1}))^T \Sigma_{\theta}^{v^{-1}}(z_{t-1}) (X_t - \mu_{\theta}^v(z_{t-1}))$$

$$+ \sum_{u < v} A_t^{uv} \log \pi_{\theta}^{uv}(z_{t-1}) + (1-A_t^{uv}) \log(1 - \pi_{\theta}^{uv}(z_{t-1}))$$

3.

$$X_t, A_t = f_{\theta}(z_{t-1}), g_{\theta}(z_{t-1}) \Rightarrow p_{\theta}(X_t, A_t | z_{t-1}) = \delta(X_t - f_{\theta}(z_{t-1})) \delta(A_t - g_{\theta}(z_{t-1}))$$

$$\Rightarrow \text{ELBO}(X_t, A_t; \theta, \phi) = -\infty$$

in the expectation all $p_{\theta}(X_t, A_t | z_{t-1})$ become zero.

multimodality:

stochastic decoder had the ability to map shared information to different classes or modes.

Deterministic decoder can only cover a single output given z .

stability:

ELBO is $-\infty$, learning becomes unstable and difficult.

$$4. \quad X_{1:T}, A_{1:T}$$

for $h \in \{1, \dots, H\}$

$$Z_{T+h-1} \sim q_\phi(Z_{T+h-1} | X_{1:T+h-1}, A_{T+h-1})$$

$$X_{T+h}, A_{T+h} \sim p_\theta(X_{T+h}, A_{T+h} | Z_{T+h-1})$$

if $q_\phi(Z_+ | X_{1:t}, A_{1:t})$ is close enough to $p_\theta(Z_+ | X_{1:t}, A_{1:t})$

the distribution of generated samples converges to

+ The Markov chain must be ergodic.

5.

$q_\phi(Z_+ | X_{1:t}, A_{1:t})$ can differ from $p_\theta(Z_+ | Z_{t-1})$ even with the D_{KL} term. in generation the model sees $\tilde{G}_{1:t}$ to generate $\tilde{Z}_{t+1}$, which are self generated.

$\Rightarrow$ The difference creates an error at each step which can accumulate over time and cause distribution shift.

method:

$$Z_t^{(0)} \sim q_\phi(Z_+ | X_{1:t}, A_{1:t})$$

for $i \in \{1, \dots, K\}$

$$Z_+^{(i)} = G(Z_+^{(i-1)}, \theta, \phi) \quad \text{gradient ascent on ELBO}$$

this method pushes the Z closer to prior.

These refined Z s minimize amortization gap and decoder remains tight to prior.